

AIMS
Airbus Industrie Material Specification

Aluminium alloy (2024)
Solution treated, controlled stretched
and naturally aged (T351)
Clad sheet
 $0,8 \text{ mm} < a \leq 6,0 \text{ mm}$
Standard surface quality
Chemical milling quality

Material Specification

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0 Introduction

For the use of this material specification, see EN 4500-2.

1 Scope

This specification defines the requirements of aluminium alloy 2024 T351 clad sheet $0,8 \text{ mm} < a \leq 6,0 \text{ mm}$ standard surface quality, for aerospace application. The product is suitable for chemical milling.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Normative references

This Airbus Industrie specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus Industrie specification only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred shall be applied.

AIMS 03-04-000 :	Technical specification - Aluminium and aluminium alloy wrought products - Sheet, plate and strip up to 12 mm.
EN 2032-2 :	Aerospace series - Metallic materials - Part 2 Coding of the metallurgical condition in delivery condition
EN 4500-2 :	Aerospace series - Metallic materials. Rules for drafting and presentation of material standards. Part 2: Specific rules for aluminium, aluminium alloys and magnesium alloys.

3 Technical requirements

Specification values are minimum except where stated.

1	Material designation		Aluminium alloy 2024										
2	Chemical composition %	Element	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	others		Al
											each	total	
		min.	-	-	3,8	0,30	1,2	-	-	-	-	-	-
max.	0,50	0,50	4,9	0,9	1,8	0,10	0,25	0,15	0,05	0,15			
3	Method of melting		-										
4	Form		clad sheet										
	Method of production		Rolled										
	Limit dimension (mm)		$0,8 < a \leq 6,0$										
5	Technical specification		AIMS 03-04-000										

6	6.1 Delivery condition and heat treatment		T351 Solution treated 490°C ≤ Θ ≤ 500°C/WQ Θ ≤ 40°C + controlled stretched (0,5 - 2,0 %) and naturally aged Θ = ambient/t≥5d										
	Delivery condition code		U ¹⁾										
7	Use condition and heat treatment		T351 as delivered										

8	Test sample Test piece Heat treatment			T351						
9	Dimensions concerned	a	mm	0,8 < a ≤ 1,6		1,6 < a ≤ 3,2		3,2 < a ≤ 6,0		
10	Thickness of cladding on each face		%	4		2		2		
11	Direction of test piece			L	LT	L	LT	L	LT	
12	T	Temperature	Θ °C	Room temperature						
13		Proof stress	R _{p0,2} MPa	305	270	310	280	310	280	
14		Strength	R _m MPa	415	410	430	425	435	435	
15		Elongation	A ₅₀ %	-	15	-	15	-	15	
16		Reduction of area	Z %							
17	Hardness		HB -							
18	Shear strength		R _c MPa							
19	Bending		k -	2,5		3		4		
20	Impact strength			-						
21	C	Temperature	Θ °C	-						
22		Time		h	-					
23		Stress	σ _a MPa	-						
24		Elongation	a %	-						
25		Rupture stress	σ _R MPa	-						
26		Elongation at rupture	A %	-						
27	Notes (see line 98)			1) 2)						

38	Intergranular corrosion		1	See AIMS 03-04-000							
			7	Thickness (mm)		0,8 < a ≤ 3,2			3,2 < a ≤ 6,0		
				max. depth of attack (µm)		See Note ²⁾			200		
44	External defects		-	See AIMS 03-04-000 Table 6							
57	Residual stress		1	See AIMS 03-04-000 Figure 4							
			7	Thickness(mm)	0,8 < a ≤ 1,0	1,0 < a ≤ 1,2	1,2 < a ≤ 1,6	1,6 < a ≤ 6,0			
				h _{max} (mm)	≤ 8	≤ 7	≤ 5	≤ 3,2			
62	Diffusion in the cladding		1	See AIMS 03-04-000							
			7	Thickness (mm)	0,8 < a < 1,0	1,0 ≤ a < 1,2	1,2 ≤ a ≤ 1,6	1,6 < a < 2,0			
				Allowable diffusion depth of actual cladding thickness in %	45	35	30	35			
			7	Thickness (mm)	2,0 ≤ a < 2,5	2,5 ≤ a < 3,2	3,2 ≤ a ≤ 6,0				
				Allowable diffusion depth of actual cladding thickness in %	30	25	20				
			64	Surface condition roughness		1	See AIMS 03-04-000				
7	Thickness (mm)	0,8 < a ≤ 4,0				4,0 < a ≤ 6,0					
	Mean roughness value Ra (µm)	4,0				6,3					
72	Cladding chemical composition	Element	Si + Fe	Cu	Mn	Mg	Zn	Ti	others		Al
									each	total	
		min.	-	-	-	-	-	-	-	-	Base
		max.	0,7	0,10	0,05	0,05	0,10	0,03	0,03	-	
82	Batch uniformity		1	See AIMS 03-04-000							
			7	Thickness (mm)		a ≥ 1,6					
				Electrical conductivity		γ = 17,5 MS/m (typical value)					
				or							
				Hardness		100 HB (typical value)					
98	Notes		-	1) See EN 2032-2							
				2) An intergranular attack like Figure 10 and 11 in AIMS 03-04-000 continuously over the entire surface of the metallographic sample, whatever depth, is not allowed. The sample for the intergranular test shall be taken from a location within the product representing maximum quench delay during heat treatment.							
100	Qualification			Frequency of qualification testing is included in AIMS 03-04-000 Recommended thicknesses are 1.0; 1.6 and 4.0 mm							

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